Alluvial ridge construction and channel–floodplain connectivity: an independent numerical reconstruction

Independent implementation of the written model specification of Nicholas et al. (2018)

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Abstract

The balance between channel bed aggradation and overbank deposition controls channel depth and the pathways by which water reaches a floodplain. Nicholas et al. [1] used this balance to relate alluvial ridge relief to channel–floodplain connectivity over 580 repeated floods. This report documents an independent reconstruction from the paper and its written supplement. It records the physical specification, numerical assumptions, verification evidence, and the status of all 13 manuscript panels. Numerical panels must be generated from saved output of this implementation; published curves and raster values are not model or plotting inputs. The current completion status and remaining limitations are reported explicitly below.

1 Scope and current evidence

The calculation tests whether a coupled one-dimensional channel and two-dimensional floodplain model yields the spatial and temporal behavior reported by Nicholas et al. [1]. Their calculations describe the period preceding an avulsion. Neither their prescribed channel migration model nor the reconstruction should be interpreted as a prediction of an avulsion occurrence or recurrence interval.

The permitted scientific inputs are published descriptions and configuration values. The authors' original implementation, simulation files, plotting scripts, and source data are excluded. Published figures were inspected for comparison, without digitization. Published values in the supplement's results columns are not substitutes for computed values. No published figure is embedded in this report. There are no supplementary figures in GSA Data Repository 2018219; DR1 is a written model description and DR2 is a parameter and results table [2].

Current panel ledger: 13 not reproduced. Panels marked partial or not reproduced have missing outputs or agreement limits described in the ledger. Rebuild execution is recorded separately. This report merges the maintained scientific ledger with current plotted-data coverage. Complete numerical coverage alone is classified as partial until a scientific assessment is documented and bound to the current numerical inputs, processors and figure. Source-ledger notes are retained as prior documentation, and current coverage and provenance are listed separately for every panel.

2 Physical specification

We model a floodplain that is 48 km wide and 72 km long. The documented grid spacing is 250 m, giving 192 by 288 cells. The initial floodplain is planar with downstream slope 10^{-4}. We set its initial elevation to $z_0(x) = 10^{-4}(72\,000\,\mathrm{m} - x)$, where x increases downstream from the inlet. This vertical

datum is an implementation convention. The initially straight channel has width $B = 500$ m and depth 8 m. A flood lasts $T_f = 20$ days and has minimum and peak discharges of 2,500 and 15,000 $\mathrm{m^3\ s^{-1}}$. Each configuration requires 580 floods with the same hydrograph. These values are specified by DR1 [2]; the hydrograph shape is not specified there.

The channel bed rises by a prescribed increment A after each flood. At any given flood it has constant slope in channel arc length s. For a channel of length L_c, the slope is $7.2\,\mathrm{m}/L_c$, so the inlet–outlet bed elevation drop remains 7.2 m as sinuosity changes. For flood index $n \geq 1$, the bed during the hydrograph is

$$z_c(s, n) = z^0_{c,\mathrm{in}} + (n-1)A - 7.2\,\mathrm{m}\frac{s}{L_c(n)}, \tag{1}$$

where $z^0_{c,\mathrm{in}} = -0.8$ m and $L_c(n)$ is the channel length during flood n. Saved rasters, channel coordinates, and end-flood diagnostics precede that flood's migration, cutoff, and bed increment. Thus a raster labeled flood n contains $(n-1)A$ of imposed bed rise. The restart checkpoint follows migration and the nth increment. Equation (1) imposes bed elevation independently of suspended sediment exchange; it does not predict bedload transport.

The inlet concentration follows $S = CQ^{1.5}$. The supplement does not state units for S or C. Any conversion to SI concentration must be identified as an assumption. The default grain fractions are 5% sand, 75% silt, and 20% clay; the fine mixture is 5% sand, 20% silt, and 75% clay. The bank erodibility multiplier E is usually 1, with selected values of 0.5 and 1.5. Appendix B lists all 31 configurations.

2.1 Conservation laws and coupling

The written specification requires depth-averaged shallow-water flow and three suspended sediment fractions. In a floodplain cell, let h be water depth, $\mathbf{u} = (u, v)$ velocity, z bed elevation, and c_k the suspended mass concentration for grain class k. Away from external boundaries, the governing balances have the form

$$\partial_t h + \nabla \cdot (h\mathbf{u}) = 0, \tag{2}$$

$$\partial_t(h\mathbf{u}) + \nabla \cdot \left(h\mathbf{u} \otimes \mathbf{u} + \tfrac{1}{2}gh^2\mathbf{I}\right) = -gh\nabla z - \mathbf{f}_b + \nabla \cdot \mathbf{T}, \tag{3}$$

$$\partial_t(hc_k) + \nabla \cdot (h\mathbf{u}c_k) = \nabla \cdot (hD_k\nabla c_k) - d_k. \tag{4}$$

Here $\mathbf{f}_b$ is bed resistance per unit water density, $\mathbf{T}$ represents any chosen horizontal momentum diffusion, D_k is sediment diffusivity, and d_k is deposition per unit bed area and time. The written sources determine the required transport structure; the final constitutive choices and any omitted terms must be stated in the implementation record below. If concentration is in kg $\mathrm{m^{-3}}$, d_k is in kg $\mathrm{m^{-2}\ s^{-1}}$. Deposit thickness is related to deposited mass by

$$\partial_t z = \frac{\sum_k d_k}{\rho_s(1-p)}, \tag{5}$$

where ρ_s is sediment density and p is deposit porosity. Equation (5) applies to the modeled depositional bed changes, not to the separately prescribed channel aggradation.

The channel uses section-integrated water and momentum equations and advection–diffusion of three suspended fractions, with lateral exchange and boundary supply. Net exchange between channel suspension and its prescribed bed is set to zero. Channel aggradation and geometry adjustments are independent signed bed forcing, recorded separately in the solid-mass budget. DR1 prescribes the bed and excludes bedload-based updates, but does not establish zero net exchange with suspension [2]. This is therefore a physical assumption, not a recovered closure. Channel variables are interpolated to

floodplain-grid cells within $B/2$ of the centerline, with velocity rotated into the local channel tangent. Interface face fluxes update the floodplain and the corresponding channel section with equal and opposite water and sediment mass transfers. The overlaid channel cells must not duplicate the water or sediment storage already represented by the one-dimensional channel.

At the end of each flood, the written specification calls for Howard–Knutson migration with neck cutoffs [3]. Newly occupied channel cells are lowered to the interpolated channel bed. Their eroded sediment is returned at the upstream concentration boundary. The timing, grain allocation, and mass accounting for this return are implementation choices where DR1 is incomplete. The topography of abandoned channel cells, cutoff handling, and remapping of water and sediment also require explicit choices. A migration displacement is not itself a shallow-water avulsion.

2.2 Implementation and unresolved assumptions

The computational model is C++17. It uses the specified 250 m floodplain grid and rectangular channel sections at approximately 250 m spacing. The channel mask consists of grid centers within 250 m of the centerline. The nearest segment gives linear interpolation weights, which are also used for incoming exchange. An outgoing mass is removed in proportion to each section's contribution to the interpolated donor inventory; this prevents withdrawal from an empty section. The channel's exact rectangular control areas replace masked floodplain storage. The raster bed inventory and the rectangular water storage consequently have slightly different footprints.

We inherit channel and floodplain Chezy coefficients of 55 and 10 $\mathrm{m}^{1/2}\ \mathrm{s}^{-1}$ from Nicholas et al. (2013) [5]. Let C_h denote this coefficient, distinct from sediment coefficient C. The resistance and eddy viscosity are

$$\mathbf{f}_b = \frac{g}{C_h^2}\mathbf{u}|\mathbf{u}|, \qquad u_* = \frac{\sqrt{g}}{C_h}|\mathbf{u}|, \qquad \nu_t = 0.07 u_* h. \tag{6}$$

The symmetric velocity-gradient tensor gives horizontal turbulent stress; we set sediment diffusivity equal to ν_t, an assumed Schmidt number of one. A local friction update follows each transport substep.

We assume that S is in mg L^{-1}, consistent with concentration units reported in the 2013 methods paper, so internal inlet concentration is $10^{-3} C Q^{3/2}$ kg m^{-3}. At the peak discharge, $C = 0.003$ gives 5.511 kg m^{-3}. The 2018 paper does not establish this unit convention. Porosity 0.4 and grain density 2650 kg m^{-3} are assumed. Sand settles at 0.023244 m s^{-1}, calculated from the Ferguson–Church relation [6] for assumed 0.2 mm particles. Silt uses 0.0002 m s^{-1} from the earlier methods paper. Clay uses Stokes settling for assumed unflocculated 2 μm spheres, giving 3.597×10^{-6} m s^{-1}. For silt and clay, the deposition sink is

$$d_k = w_{s,k} c_k \max(0, 1 - \tau_b/\tau_c), \qquad \tau_b = \rho_w g |\mathbf{u}|^2 / C_h^2, \tag{7}$$

with assumed $\tau_c = 0.1$ Pa. Sand deposition is independent of this shear reduction. At fixed depth and stress, deposition integrates exponentially, and the removed suspended mass increments bed elevation exactly once. Vertical floodplain entrainment is omitted because the earlier sand-capacity-based closure does not close the three-fraction, prescribed-channel-bed problem. This omission is a physical uncertainty.

The floodplain is initially dry. The channel starts at the minimum-discharge normal depth, 4.356 m, with zero suspended concentration. The unreported hydrograph shape is taken as

$$Q(t) = 2500 + 12500 \sin^2(\pi t / T_f) \quad \mathrm{m}^3\,\mathrm{s}^{-1}. \tag{8}$$

Water and sediment fields persist between floods. At the inlet we prescribe Q and preserve the outgoing shallow-water characteristic $u - 2\sqrt{gh}$. At the subcritical outlet we preserve $u + 2\sqrt{gh}$ and impose the

Chezy normal-flow rating for the current channel slope. These boundary conditions are assumptions. Unsupported incoming supercritical states stop the calculation with an explicit error. Floodplain lateral and upstream boundaries reflect flow; the downstream boundary is transmissive with continued valley slope.

Floodplain transport rates are evaluated once per macro step, with CFL coefficient 0.4, a 120 s ceiling, and positive-inventory restrictions. The channel takes additional wave-limited substeps with fixed lateral exchange rates during that interval. The scheme is first order in time. Both representations receive identical integrated water and sediment exchange; channel momentum exchange includes the advective streamwise component. Side pressure is normal to the physical streamline and is not added as a streamwise force from raster stair steps. Exactly dry regions are skipped computationally without changing active-state arithmetic.

Migration uses the linear-curvature Howard–Knutson/Ikeda parameterization written in later methods descriptions [7, 8]. For curvature κ, define

$$R_0 = k_l B \kappa, \qquad R_1 = E\left[-R_0 + 2.5\frac{\int_0^s R_0(\xi)e^{-a(s-\xi)}\,d\xi}{\int_0^s e^{-a(s-\xi)}\,d\xi}\right], \quad a = \frac{2g}{C_h^2 H}. \tag{9}$$

We use assumed $k_l = 400$ m per flood, bankfull depth H, fixed endpoints, and 5 m initial random perturbations with seed 2018. Here k_l is the coefficient per unit dimensionless curvature, not the mean observed migration speed. The upstream integral is exact for piecewise-linear curvature. Adaptive steps and geometric resampling control numerical crowding. Nonadjacent limbs closer than one channel width form a neck cutoff. The precise 2018 migration parameterization is unverified.

Migration precedes prescribed aggradation. Newly cut cells are excavated to the pre-aggradation channel bed; abandoned cells retain their former bed. Eroded dry mass enters a reservoir and is returned at the next inlet with the selected grain fractions and a $Q^{3/2}$ concentration shape. Release is limited by available reservoir inventory, with quadrature remainders carried forward. To transfer abandoned cells, we normalize the old interpolation weights over each section's entire raster support. If $V_j q_j$ is its inventory and $S_j = \sum_r a_r w_{rj}$ its sampled area, abandoned cell r receives $\sum_j (a_r w_{rj}/S_j) V_j q_j$. The retained support fraction stays in the channel; unsupported sections retain all their inventory. Conservative overlap in normalized arc length remaps the remaining channel state. Newly captured floodplain inventories enter neighboring new sections using the new interpolation weights. Water and every sediment class share these nonnegative weights, preserving mass and constant concentrations. Local depth need not be preserved when raster and rectangular areas differ. The earlier subtraction of full ghost-cell inventories failed a 250 m cutoff test by overdrawing a localized sediment class by 35.48%; that attempt was stopped and its records retained. Abandoned momentum rotates into the old segment tangent; incoming floodplain momentum projects onto the new segment tangent. Geometry change therefore does not conserve Cartesian momentum. This remapping closure and fixed bank grain fractions are explicit assumptions. Prescribed bed changes have a separate signed sediment source in the mass ledger. A nonpositive bankfull depth is treated as a failure of the prescribed channel geometry, not replaced by an artificial minimum.

For α, the floodplain depth average includes dry cells. A bank face is inundated when floodplain water and channel overtopping both exceed 1 mm; this threshold affects diagnostics only. No wet interface gives an undefined value. The immediate active-channel interface is consistent with visual inspection of published Figure 1E, although the raster implementation and wetness criterion are assumptions. Ridge relief for β uses the maximum transverse value of longitudinal-mean relief over the central 80% of the reach, relative to the outermost 10% lateral bands. Channel depth is the bank-face mean elevation minus adjacent channel bed. We average the resulting β_n over floods 561–580. These averaging choices are not specified in DR1.

The horizontal Boussinesq turbulent stress is implemented, but the secondary-flow and associated dispersive terms in the earlier methods paper are omitted [5]. These terms describe effects of unresolved vertical velocity structure; horizontal eddy viscosity does not replace them. Bed composition is also not tracked as deposits accumulate and are reworked. Excavated bank mass is assigned the selected inlet grain fractions. This preserves total solid mass while losing the grain identity of earlier deposits. Together with omitted vertical floodplain entrainment and zero net channel suspended-bed exchange, these choices limit correspondence with the inherited process equations.

DR1 specifies first-order spatial accuracy and an explicit finite-volume approximate Riemann solver. The independent kernel uses hydrostatic reconstruction [4]. At an interface, $z_* = \max(z_L, z_R)$ and $h_* = \max(0, h + z - z_*)$. Velocities and concentrations are retained in the reconstructed wet states. Side-specific pressure corrections balance a stationary free surface over bed steps. The HLL water flux and the sediment flux formed from upwind concentration are shared between the two sides. These are independently selected numerical details, not a claim to match the original implementation.

3 Diagnostics and verification

The published super-elevation ratio is

$$\beta_n = \frac{z_{\mathrm{ridge},n} - z_{\mathrm{distal},n}}{H_{\mathrm{channel},n}}. \tag{10}$$

The reported late-time statistic is the arithmetic mean of the 20 per-flood ratios β_n at floods 561–580. Ridge relief is the maximum transverse value of longitudinal-mean relief over the central 80% of the reach, relative to the outermost 10% lateral bands. Channel depth is the mean bank-face elevation minus adjacent channel-bed elevation. These spatial averages and the mean-of-ratios convention are implementation choices not resolved by the supplement. A constant vertical datum cancels from the numerator, but changes in the selected ridge or distal region do not.

For each flood, the connectivity measure is sampled at $t = 0.6T_f = 12$ days:

$$\alpha_n = \frac{\overline{h}_{\mathrm{fp},n}}{f_{\mathrm{wet},n}}, \qquad \alpha_{95} = P_{95}\{\alpha_{381}, \ldots, \alpha_{580}\}. \tag{11}$$

Here f_{wet} is the inundated fraction of the channel–floodplain interface. Unlike β, α has units of meters. The depth average includes every floodplain cell, including dry cells. The interface consists of internal Cartesian faces between active channel and floodplain cells; all faces have equal length at the uniform grid spacing. A face is inundated when floodplain depth and channel overtopping depth both exceed 1 mm. Zero inundated fraction gives undefined α. The percentile uses linear interpolation and requires all 200 values to be finite; missing or undefined values are neither replaced by zero nor discarded.

At day 12 the output retains α, mean floodplain depth, and inundated interface fraction for every flood. For the requested map and profile floods it also saves rasters and channel coordinates with a `peak_` prefix, at cumulative time $(n - 0.4)T_f$. These files support assessment of inundated area and breach location at the sampling time, which deposit footprint alone cannot supply. The earlier interrupted rebuild did not save these fields. Their availability and spatial interpretation remain to be assessed from the new calculations. Channel geometry is unchanged within each hydrograph, so the end-flood outline identifies the full day-12 interface, without identifying its inundated subset.

Verification must distinguish discrete conservation from agreement with the paper. Required checks include a lake at rest, positive wetting and drying, coupled water and grain-mass exchange, analytical transport and settling, zero forcing limits, and cutoff/remapping budgets. External boundary fluxes,

bed changes, and prescribed channel aggradation must be reported separately. Representative coupled cases require grid and time step sensitivity. Kernel tests alone cannot establish these properties for the full coupled model.

The retained development validation record reports passing component and coupled tests. The lake-at-rest kernel error is 6.76×10^{-15} after 1000 steps. Analytical settling conserves suspended plus deposited mass, and periodic sediment diffusion errors decrease from 1.04×10^{-5} to 3.04×10^{-6} to 7.61×10^{-7} under successive refinement. Migration tests verify curvature, the upstream convolution, symmetry, cutoff length accounting, and finite evolution over 580 migration steps. These are tests of the migration component, not 580 coupled flood runs.

Coupled tests verify reflecting boundaries, two-way class and water exchange, remapping with and without aggradation, and the complete suspension–bed–bank-reservoir ledger. A localized suspended plume remains nonnegative through 150 macro steps. Its water and sediment budget residuals are approximately 4×10^{-9} m^3 and 4×10^{-11} kg. A steady normal-channel test has maximum relative depth errors 0.179% and 0.0843% at 250 and 125 m section spacing. The imposed reservoir-release waveform error decreases from 0.004915 to 0.002461 when the CFL coefficient is halved. Positive integration remainders remain in the reservoir; injected plus remaining mass is conserved.

A two-flood short-domain checkpoint test reproduces uninterrupted raster output, channel coordinates, and the final checkpoint byte-for-byte. Changed checkpoint parameters are rejected, and diagnostic rows written after the last valid checkpoint are regenerated. These checks establish restart behavior for the tested case.

This development summary predates completion of the full-domain first-flood cases and is not a live record of their status. The separately archived findings from corrected rebuild 002 appear in Section 3.1, including completed 250 m and 500 m sensitivity cases and the captured first flood of all 31 production cases. The component tests summarized here do not establish grid independence or long-term quantitative agreement for those production configurations. Exact commands and logs are in `validation/` and `docs/numerical-validation.md`. Existing kernel verification log (scope limited to the named checks). Display lines are wrapped; the original bytes remain in `validation/kernel.txt`.

```
mkdir -p build
g++ -O3 -std=c++17 -Wall -Wextra -Wpedantic -Isrc tests/finite_volume.cpp -o
  build/finite_volume_test
./build/finite_volume_test
lake_at_rest_max_error 6.75848e-15
shared_face_water_sediment_conservation pass
analytical_settling_and_mass_conservation pass
dry_interface pass
```

The first clean rebuild was interrupted on 9 October 2026 before any production flood completed. A separate 250 m neck-cutoff test, using its frozen headers, showed that the remap could overdraw a channel section's suspended inventory by 35.48%. This verification failure prompted the interruption; it was not a remap failure observed in a production flood. The original passing checks had not exercised this counterexample. The subsequent conservative transfer correction is described above, and rebuild 002 uses the corrected implementation. The rejected attempt's unchanged sources, executable, logs, and interrupted case records are retained separately in `validation/rejected-rebuild-001/`. Its empty production summaries supply no manuscript values.

A later migration-component check tests growth and downstream phase motion of small sinusoidal perturbations in the independently implemented law. With the default migration parameters, the infinite-domain linear limit has a neutral wavelength of 6.328 km: the tested 12 km mode grows and the 4 km mode decays. At node spacings of 500, 250, and 125 m, native one-flood complex-rate errors decrease by factors 3.94–3.98. This is combined spatial and temporal refinement because the internal time-step limit scales with the square of node spacing. Controlled 0.1-flood evolutions agree with the

independently derived discrete Euler response within 6.08×10^{-9} per flood. Separate zero-duration resampling tests expose interpolation attenuation: when a nominally 250 m grid gains one node, amplitudes decrease by 0.206% and 1.848% for the 12 and 4 km modes. The 72-case record is retained in `validation/migration-linear/`, with equations and controls in `docs/migration.md`. It was run after rebuild 002 began and does not establish coupled grid convergence, late planform agreement, or the cause of the entire early production sinuosity change.

Production conservation and sensitivity require assessment over their recorded intervals; archived first-flood findings are reported below. Finite residuals alone do not establish negligible budget drift. The final assessment requires maximum and terminal residuals with stated normalization, together with the numerical field differences and diagnostic changes from grid, time-step, and physical-assumption cases. Per-flood budget summaries precede migration and aggradation. A separate `remap_budget.csv` records local water and suspended-class transfer defects, the combined solid defect after signed bed forcing, and cumulative residuals after each remap. Both cumulative ledgers reset their reference inventory at process restart; they do not recover accumulated full-history drift across restarts. The sensitivity cases cover the first flood and retain 250 m channel spacing, so they cannot establish 580-flood migration or grid independence.

3.1 Archived first-flood validation

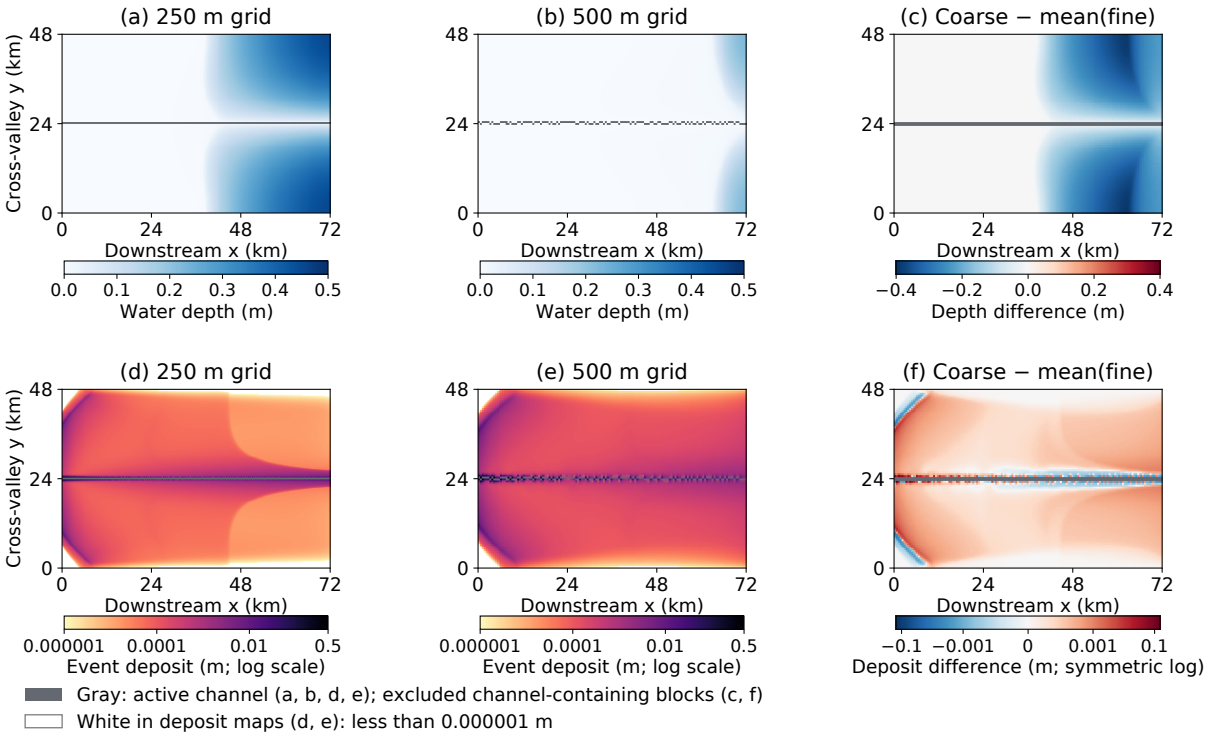
The corrected rebuild 002 provides provisional evidence over the first 20-day hydrograph. These archived observations establish a limited numerical check; they do not supply any requested manuscript panel. The 250 m and 500 m floodplain cases use configuration 11 and the full 72 km by 48 km domain. Their recorded parameters differ only in floodplain spacing; channel-node spacing remains 250 m. The capture verified both completed process records and their output hashes.

Quantity	250 m grid	500 m grid
Day-12 connectivity α (m)	8.84486×10^{-1}	7.7058×10^{-1}
End-flood ridge height (m)	1.31874×10^{-2}	3.35724×10^{-2}
End-flood relative relief β	1.64372×10^{-3}	4.17848×10^{-3}
Deposited mineral mass (kg)	2.24279×10^{9}	4.4543×10^{9}
New channel cells after remap	0	49
Bank reservoir after remap (kg)	0	1.568×10^{11}

Deposited mass is the accumulated deposition source ledger retained in the checkpoint, not a post-remap integral of the remaining deposit. The coarse-grid deposited mass exceeds the fine-grid value by 98.61%. Both day-12 inundated interface fractions equal one. The end-flood comparison averages each two-by-two block of fine cells onto the coarse grid and excludes blocks containing any channel cell. Over the resulting 3384 km^2 common floodplain, water-depth differences have mean absolute value 0.100738 m and maximum 0.386764 m. These averages include dry areas and omit the excluded channel blocks.

Validation figure V1 maps these end-flood fields and their differences.

The coarser grid captures 49 bank cells after the first hydrograph, whereas the finer grid captures none. This difference changes the reservoir available for later recycling. Capture occurs after the hydrograph and therefore cannot explain the first-flood deposit-mass difference. The adaptive time steps also change with floodplain spacing. Two first-flood cases with fixed channel spacing do not establish spatial or temporal convergence, nor do they test recycling through a subsequent complete hydrograph.



Validation figure V1. Independent first-flood calculation for configuration 11 over the 72 km by 48 km domain. Rows show day-20 water depth and event-deposit thickness before migration and remapping. The first two columns use 250 m and 500 m floodplain grids; both retain 250 m channel-node spacing. The third column subtracts the arithmetic mean of each two-by-two fine block from the coarse cell. Gray marks active channel cells, or excluded comparison blocks containing any channel cell in either grid. Dry cells remain in the comparisons. Depth scales are linear; deposit scales are logarithmic from 10^{-6} to 0.5 m, with smaller deposits white. Signed deposit differences use a base-10 symmetric logarithmic scale with a linear core of $\pm 10^{-4}$ m. Display thresholds do not modify the reported differences. These two first-flood cases establish neither convergence nor reproduction of a manuscript panel.

The historical production observation at 2026-10-09T21:39:16.936345Z records all 31 cases after their first remap, while their live progress records indicated flood 2. The recorded process identities, finite-value checks, positivity checks, and checkpoint signatures passed. Every case had zero captured or abandoned raster cells, bank reservoir, recycled mass, cutoff count, and water or mass correction ledger. The largest absolute post-remap residuals across those cases were:

Inventory	Absolute residual	Run	Relative to input	Run
Water	1.31249×10^{-4} m^3	23	8.68052×10^{-15}	23
Suspension + deposited sediment	2.51689×10^{0} kg	18	1.47412×10^{-11}	27
Full solid inventory	3.96595×10^{-1} kg	02	7.24578×10^{-10}	02

Absolute and normalized maxima are selected separately and can occur in different runs. Normalization uses the prescribed cumulative upstream water or sediment input through the first hydrograph. It excludes recycled bank material and independent signed bed forcing from the denominator; it is neither current storage nor net input. The full-solid residual includes bank and bed accounting with the prescribed signed bed forcing subtracted. These small first-flood residuals constrain conservation over the captured interval only. The separately recorded signed bed forcing ranges from -7.75454×10^5 kg (run 28) to 2.06082×10^9 kg (run 22). This prescribed geometry and aggradation forcing contributes to solid-inventory change independently of suspended transport.

The earlier coarse-grid observations remain in the same archive. This section verifies the capture manifest, archived analysis scripts and JSON records, together with all 289 retained production-input and 24 completed sensitivity-input file hashes. It reads no live process or checkpoint. Artifact and generator identities are recorded in `report/generated/first_flood_evidence.json`; the retained sources are in `validation/first-flood-audit-002`. Later conservation drift, bank recycling, finer-grid sensitivity, and long-term morphology require further assessment.

Sensitivity to diagnostic definitions. The published description does not fix all spatial averages in β. Reprocessing the same archived day-20 fields gives:

Nominal central reach	β, 250 m	Change (%)	β, 500 m	Change (%)
60%	1.287×10^{-3}	-21.73	3.426×10^{-3}	-18.02
80%	1.644×10^{-3}	0.00	4.178×10^{-3}	0.00
100%	2.852×10^{-3}	73.52	5.602×10^{-3}	34.07

Only the reach used for the transverse relief profile changes in this table. The distal bands remain at nominal 10% of the domain width at each side, and the denominator remains the full-interface mean bankfull depth. Changes are relative to the raster reconstruction of the default central-80% estimator. Integer trimming gives central lengths of 43.5 and 44 km at nominal 60%, respectively; both grids use 58 km at nominal 80% and 72 km at 100%. Their default distal bands are 4.75 and 4.5 km wide on each side. Thus including the end regions materially changes first-flood ridge relief. This comparison identifies sensitivity to the averaging definition; it does not isolate a boundary-condition mechanism.

At the default reach, changing each distal band to 5%, 10%, or 20% changes β by at most 0.24%. Holding the default ridge relief fixed and restricting the channel-depth average to bank faces in the nominal central 60%, 80%, or full reach changes β by at most 0.20%. The source record retains all nine reach/band combinations and the three separate denominator variants.

For day-12 α, thresholds of 0.1, 1, and 10 mm leave every sampled bank face wet in both cases. The full-domain mean depth, including dry cells, is unchanged, so α is unchanged. The nominal wet floodplain

area nevertheless varies with the threshold: 3389.625–3404.875 km^2 at 250 m and 3371.250–3406.750 km^2 at 500 m. No tested wetness classification is ambiguous within float32 rounding.

The default reconstructions of ridge relief, channel depth, β, and α agree with the double-precision summaries within propagated half-ULP snapshot bounds and the recorded arithmetic and CSV rounding allowances. These bounds describe representation precision, not discretization or physical uncertainty. The scenarios do not select a preferred estimator and do not establish robustness of the late 20-flood or 200-flood manuscript statistics. `figures/first_flood_diagnostic_sensitivity.json` records the definitions, effective grid windows, bounds, and input identities.

3.2 Completed first-flood time-step sensitivity

A later completed case reduces the CFL limit from 0.4 to 0.2 while holding the corrected executable, configuration 11, 250 m floodplain grid, and 250 m initial channel spacing fixed. Both calculations complete the full 20-day hydrograph with exit code zero. The retained archive contains both original process records and all recorded outputs. Macrosteps increase from 125,765 to 250,265 and channel substeps from 452,282 to 899,949; adaptive steps are not uniformly halved at every time.

Quantity	CFL 0.4	CFL 0.2	Change (%)
Day-12 connectivity α (m)	8.84486×10^{-1}	8.77252×10^{-1}	-0.81785
End-flood ridge height (m)	1.31874×10^{-2}	1.32257×10^{-2}	+0.28975
End-flood relative relief β	1.64372×10^{-3}	1.64847×10^{-3}	+0.28894
Mean bankfull channel depth (m)	8.02294×10^{0}	8.02301×10^{0}	+0.00081
End-flood water storage (m^3)	5.55865×10^{8}	5.22785×10^{8}	-5.95097
Deposited mineral mass (kg)	2.24279×10^{9}	2.07035×10^{9}	-7.68847

Percentage changes use the CFL 0.4 value as denominator. Connectivity is sampled on day 12; ridge height, relative relief, channel depth, and water storage are sampled on day 20 before migration/remapping. Deposited mineral mass is the accumulated source ledger retained in the post-remap checkpoint, not an integral of the remaining deposit. Both inundated interface fractions equal one, so the connectivity change is the change in mean floodplain depth.

The end-flood masks agree. Over all 3420 km^2 of floodplain, including dry cells, water-depth differences have mean absolute value 0.00941692 m and maximum 0.06339424 m. Event-deposit thickness differences have mean absolute value 0.00003672 m and maximum 0.00362724 m. The respective bounds from float32 storage on these mean differences are 9.25765×10^{-9} m and 3.50178×10^{-11} m; they exclude discretization and separate floating-point arithmetic error.

The half-CFL post-remap residuals are -1.71661×10^{-5} m^3 water, -1.46355×10^{0} kg suspension plus deposition, and -6.92458×10^{-1} kg total solids. Normalized by prescribed upstream inputs of 1.512×10^{10} m^3 and 5.47346×10^{10} kg, their magnitudes are below 2.7×10^{-11}. These denominators exclude recycling, outflow, and signed bed forcing. Both first remaps have zero captured or abandoned cells, bank reservoir, recycled mass, cutoffs, and water or mass corrections. Signed bed forcing is retained as a separate ledger in the comparison.

The distinct sensitivities of water storage and ridge height require both hydraulic and deposition quantities to be assessed. Two CFL limits do not determine a temporal convergence order or establish time-step independence. This first-flood comparison does not test later recycling, 580-flood evolution, or manuscript-panel agreement. The portable replay command is:

```
python3 -B validation/first-flood-audit-002/time-step.py
```

3.3 Completed first-flood deposition-threshold sensitivity

This comparison varies the assumed critical deposition stress τ_c from a 0.1 Pa baseline to 0.05 and 0.2 Pa. The silt and clay settling rates contain the factor $\max(0, 1 - \tau_b/\tau_c)$, where τ_b is bed shear stress. At fixed shear stress and concentration, lowering τ_c suppresses settling; raising it enhances settling where the shear reduction is active. Sand settling has no shear reduction. All three cases use the corrected executable, configuration 11, 250 m floodplain grid, 250 m initial channel spacing, CFL limit 0.4, and one complete 20-day hydrograph. All other numerical parameters agree. All three recorded processes exit successfully.

Quantity	$\tau_c = 0.1$ Pa	$\tau_c = 0.05$ Pa	$\tau_c = 0.2$ Pa
Day-12 connectivity α (m)	8.84486×10^{-1}	8.8449×10^{-1}	8.84476×10^{-1}
End-flood ridge height (m)	1.31874×10^{-2}	1.41532×10^{-2}	1.23206×10^{-2}
End-flood relative relief β	1.64372×10^{-3}	1.76389×10^{-3}	1.53585×10^{-3}
Mean bankfull depth (m)	8.02294×10^{0}	8.02386×10^{0}	8.02202×10^{0}
End-flood water storage (m^3)	5.55865×10^{8}	5.56111×10^{8}	5.55426×10^{8}
Deposited mineral mass (kg)	2.24279×10^{9}	1.80697×10^{9}	3.27677×10^{9}

Percentage change from 0.1 Pa	$\tau_c = 0.05$ Pa	$\tau_c = 0.2$ Pa
Day-12 connectivity α (m)	+0.00043	-0.00114
End-flood ridge height (m)	+7.32334	-6.57286
End-flood relative relief β	+7.31106	-6.56210
Mean bankfull depth (m)	+0.01145	-0.01151
End-flood water storage (m^3)	+0.04423	-0.07885
Deposited mineral mass (kg)	-19.43183	+46.10262

Changes are each candidate minus the 0.1 Pa baseline; percentages use the baseline value as denominator. Connectivity is sampled on day 12; ridge height, relative relief, channel depth, and water storage are sampled on day 20 before migration/remapping. The post-remap checkpoint ledger records mineral mass added by deposition; the raster stores remaining event-deposit thickness, including pore space. All three day-12 inundated interface fractions equal one, so the connectivity difference is a mean-depth difference.

At 0.05 Pa, ridge height changes by +7.323% and deposited mineral mass by -19.432% relative to the baseline. At 0.2 Pa, ridge height changes by -6.573% and deposited mineral mass by +46.103% relative to the baseline. Ridge relief and integrated deposition respond differently in these calculations. The local settling factor alone does not determine the sign of the coupled spatial response.

The channel/floodplain classification masks agree across all three cases at peak and end flood. The field comparisons below cover all 3420 km^2 of floodplain, including dry cells. Each difference is the candidate minus the baseline. The last column bounds float32 storage error in the mean difference; it excludes discretization, separate floating-point arithmetic error, and rounding of the tabulated values. All entries are in metres.

$\tau_c = 0.05$ Pa relative to 0.1 Pa

Phase	Field	Mean $\lvert\Delta\rvert$	Max. $\lvert\Delta\rvert$	Mean storage bound
Day 12	Water depth	3.72141×10^{-5}	7.88608×10^{-3}	7.13658×10^{-8}
Day 12	Event deposit	3.70402×10^{-5}	8.43491×10^{-3}	1.93411×10^{-11}
Day 20	Water depth	9.53963×10^{-5}	3.67419×10^{-3}	9.58587×10^{-9}
Day 20	Event deposit	1.12036×10^{-4}	8.22584×10^{-3}	3.28866×10^{-11}

$\tau_c = 0.2$ Pa relative to 0.1 Pa

Phase	Field	Mean $\lvert\Delta\rvert$	Max. $\lvert\Delta\rvert$	Mean storage bound
Day 12	Water depth	9.97063×10^{-5}	1.09373×10^{-2}	7.13673×10^{-8}
Day 12	Event deposit	9.61107×10^{-5}	1.19132×10^{-2}	2.37361×10^{-11}
Day 20	Water depth	1.60123×10^{-4}	3.18443×10^{-3}	9.57521×10^{-9}
Day 20	Event deposit	2.41958×10^{-4}	1.16961×10^{-2}	4.44829×10^{-11}

Post-remap residuals retain their signs. The water denominator is the prescribed upstream input of 1.512×10^{10} m^3; both sediment denominators are the prescribed upstream mineral input of 5.47346×10^{10} kg. These common denominators exclude recycling, outflow, and signed bed forcing; they are neither current storage nor net external input.

τ_c (Pa)	Inventory	Signed residual	Residual/input
0.1	Water (m^3)	-1.91927×10^{-5}	-1.26936×10^{-15}
0.1	Suspension + deposition (kg)	-7.54353×10^{-1}	-1.3782×10^{-11}
0.1	Full solids (kg)	-3.60248×10^{-1}	-6.58173×10^{-12}
0.05	Water (m^3)	-5.13196×10^{-5}	-3.39415×10^{-15}
0.05	Suspension + deposition (kg)	-6.3638×10^{-1}	-1.16266×10^{-11}
0.05	Full solids (kg)	-3.44117×10^{-1}	-6.28701×10^{-12}
0.2	Water (m^3)	-4.63724×10^{-5}	-3.06696×10^{-15}
0.2	Suspension + deposition (kg)	-1.09959×10^{0}	-2.00894×10^{-11}
0.2	Full solids (kg)	-3.61684×10^{-1}	-6.60797×10^{-12}

All three first remaps have zero captured or abandoned cells, bank mass, recycled mass, cutoffs, and water or mass correction terms. The signed bed-forcing ledgers are 1.76403×10^5, 1.76447×10^5, 1.76359×10^5 kg for 0.1, 0.05, and 0.2 Pa, respectively. This forcing contributes to solid-inventory change independently of suspended transport and is subtracted in the full-solid residual.

This result quantifies sensitivity to an assumed deposition parameter. It does not constrain the appropriate physical threshold, numerical convergence, later bank recycling, 580-flood behavior, or agreement with a manuscript panel. The portable replay command is:

```
python3 -B validation/first-flood-audit-002/threshold.py
```

3.4 Completed first-flood concentration sensitivity

The manuscript leaves the concentration units in $S = CQ^{1.5}$ unstated. This comparison halves and doubles the explicit multiplier that converts that law to internal mass concentration, from the 0.001 baseline to 0.0005 and 0.002, at fixed $C = 0.003$. All three cases retain configuration 11, the corrected executable, 250 m floodplain and initial channel spacing, CFL limit 0.4, and the complete 20-day hydrograph. All other numerical parameters agree, and all three recorded processes exit successfully. The retained baseline process, parameters, and remap ledger match the earlier resolution archive.

Quantity	Baseline	Half	Double
Day-12 connectivity α (m)	8.84486×10^{-1}	8.84358×10^{-1}	8.84647×10^{-1}
End-flood ridge height (m)	1.31874×10^{-2}	6.55158×10^{-3}	2.67555×10^{-2}
End-flood relative relief β	1.64372×10^{-3}	8.17777×10^{-4}	3.32525×10^{-3}
Mean bankfull depth (m)	8.02294×10^{0}	8.01145×10^{0}	8.04618×10^{0}
End-flood water storage (m^3)	5.55865×10^{8}	5.54764×10^{8}	5.57733×10^{8}
Deposited mineral mass (kg)	2.24279×10^{9}	1.12393×10^{9}	4.46263×10^{9}

Percentage change from baseline	Half	Double
Day-12 connectivity α (m)	-0.01447	+0.01817
End-flood ridge height (m)	-50.31951	+102.88661
End-flood relative relief β	-50.24825	+102.30074
Mean bankfull depth (m)	-0.14324	+0.28960
End-flood water storage (m^3)	-0.19803	+0.33618
Deposited mineral mass (kg)	-49.88691	+98.97701

Percentage changes use the baseline as denominator. Connectivity is sampled on day 12; ridge height, relative relief, channel depth, and water storage are sampled on day 20 before migration/remapping. Deposited mineral mass is the accumulated deposition source ledger in the post-remap checkpoint. It differs from the remaining porous event-deposit thickness stored in the raster. All three day-12 inundated interface fractions equal one, so each connectivity change is a mean-depth change.

The channel/floodplain classification masks agree at peak and end flood. The following differences cover all 3420 km^2 of physical floodplain, including dry cells, and subtract baseline from each candidate. All values are in metres. The storage bounds apply only to float32 representation of the mean differences; they exclude discretization, separate arithmetic error, and tabulation rounding.

Half concentration relative to baseline

Phase	Field	Mean $\lvert\Delta\rvert$	Max. $\lvert\Delta\rvert$	Mean storage bound
Day 12	Water depth	3.57009×10^{-4}	1.18062×10^{-1}	7.13556×10^{-8}
Day 12	Event deposit	1.18006×10^{-4}	1.32×10^{-1}	1.54189×10^{-11}
Day 20	Water depth	3.53956×10^{-4}	4.33312×10^{-3}	9.56831×10^{-9}
Day 20	Event deposit	2.05755×10^{-4}	1.62277×10^{-1}	2.7366×10^{-11}

Double concentration relative to baseline

Phase	Field	Mean $\lvert\Delta\rvert$	Max. $\lvert\Delta\rvert$	Mean storage bound
Day 12	Water depth	6.71152×10^{-4}	2.17374×10^{-1}	7.13884×10^{-8}
Day 12	Event deposit	2.3608×10^{-4}	2.44559×10^{-1}	3.08121×10^{-11}
Day 20	Water depth	6.86106×10^{-4}	9.28627×10^{-3}	9.60609×10^{-9}
Day 20	Event deposit	4.08299×10^{-4}	2.88753×10^{-1}	5.39156×10^{-11}

The signed post-remap residuals below come from the double-precision ledgers. All three water residuals use prescribed upstream water input of 1.512×10^{10} m^3 as denominator. The sediment denominators are 5.47346×10^{10}, 2.73673×10^{10}, 1.09469×10^{11} kg for baseline, half, and double concentration, respectively. Each is independently integrated from the case-specific multiplier times $CQ^{5/2}$ with 20,000-interval Simpson quadrature. These denominators exclude recycling, outflow, and signed bed forcing; they are neither current storage nor net external input.

Case	Inventory	Signed residual	Residual/input
Baseline	Water (m^3)	-1.91927×10^{-5}	-1.26936×10^{-15}
Baseline	Suspension + deposition (kg)	-7.54353×10^{-1}	-1.3782×10^{-11}
Baseline	Full solids (kg)	-3.60248×10^{-1}	-6.58173×10^{-12}
Half	Water (m^3)	4.69089×10^{-5}	3.10244×10^{-15}
Half	Suspension + deposition (kg)	-3.82729×10^{-1}	-1.39849×10^{-11}
Half	Full solids (kg)	-3.51605×10^{-1}	-1.28476×10^{-11}
Double	Water (m^3)	-8.16584×10^{-5}	-5.40069×10^{-15}
Double	Suspension + deposition (kg)	-1.45685×10^{0}	-1.33083×10^{-11}
Double	Full solids (kg)	-3.20001×10^{-1}	-2.92321×10^{-12}

All three first remaps have zero captured or abandoned cells, bank mass, recycled mass, cutoffs, and water or mass correction terms. The signed bed-forcing ledgers are 1.76403×10^5, 1.75855×10^5, 1.77506×10^5 kg for baseline, half, and double concentration, respectively. This forcing contributes to solid-inventory change independently of suspended transport and is subtracted in the full-solid residual.

This completed first-flood comparison quantifies sensitivity to an explicit concentration assumption. It does not determine the intended physical concentration, numerical convergence, later bank recycling, 580-flood behavior, or manuscript-panel agreement. The portable replay command is:

```
python3 -B validation/first-flood-audit-002/concentration.py
```

4 Independent figures and comparison targets

Figures 1–3 reserve all manuscript panels. A placeholder is a missing result. A generated panel must retain the configuration, sampling time, and plotted-data provenance in Appendix A. Status is assessed separately from whether a plotting script produced a file: a valid numerical panel may still disagree with the publication.

4.1 Flood deposits: manuscript panels 1A–F

The published comparison is between deposits from individual floods, not accumulated bed elevation. Panels 1A–D use $A = 0$, $C = 0.003$, and the default load at floods 61, 251, 501, and 562. Panels 1E and 1F use flood 562, with $(A, C) = (0.036, 0.003)$ and $(0, 0.0003)$, respectively. The published color scale spans 0–0.5 m, with deposits below 10^{-6} m distinguished from the active channel. This scale is a comparison reference, not a clipping rule for the computed mass budget.

The paper reports early splay deposition near migrating bend apices, progressive widening of the channel belt, and later deposition along abandoned channels. It reports reduced floodplain inundation as channel depth increases, with event-to-event fluctuations under identical imposed floods. The high-aggradation and lower-load cases retain more widespread inundation. Independent comparison should measure deposit area above 10^{-6} m, deposit volume, transverse spread, channel sinuosity, and abandoned-channel coverage where saved output supports these diagnostics. No pixel-wise match is expected without the undocumented initial migration perturbation and dimensional migration parameters.

4.2 Transverse elevation: manuscript panels 2A–D

The cross-section uses the downstream one of the two saved grid columns equidistant from the domain midpoint, at 36.125 km for the 250 m grid. Its initial floodplain elevation is 3.5875 m under the model datum. Each panel requests elevations at the end of floods 50, 225, 400, and 580, before migration and aggradation. The published panels show a growing ridge with a concave cross-sectional envelope, interrupted by active and abandoned channels. The approximate late-time ridge elevations visible in the published panels are 40–45 m (2A), 25–30 m (2B), 15–17 m (2C), and 30–33 m (2D). These are coarse visual ranges, not digitized observations. The initial mid-domain floodplain is plotted near 9 m in the publication; DR1 leaves the absolute vertical datum unspecified. The independent plots apply no datum shift. Quantitative comparisons subtract each initial plane and retain the uncertainty of the published visual range. Channel minima, ridge width, and the sign and scale of relief growth provide additional comparisons.

For each independent profile we exclude currently active-channel cells and define positive-relief weights $w_i = \max(z_i - z_0, 0)$ elsewhere. The section area above the initial plane is $B = \Delta y \sum_i w_i$; its equivalent width is B/H, where $H = \max_i(z_i - z_0)$ over non-active cells. The transverse standard deviation uses these weights and includes the within-cell variance $\Delta y^2/12$. Half-crest occupied width counts cells with relief at least $H/2$, whereas their full span includes gaps. Each flank's mean departure from its distal-to-crest endpoint chord describes broad shape; a positive departure lies above that chord. Active-channel gaps remain excluded and abandoned depressions remain in the flank calculation. These diagnostics do not define a published ridge width or a numerical test of the paper's concavity description. Measures with unresolved positive relief are not interpreted.

4.3 Super-elevation and connectivity: manuscript panels 3A–C

Panel 3A requires one late-time β value for each of the 31 configurations. The paper reports that β rises with A and, over most shared A values, decreases with increasing C. A finer load reduces ridge super-elevation. The prose reports approximately a 15% increase in β for lower erodibility and a 6.5% decrease for higher erodibility; these are comparisons to test, not coefficients to impose. Only one of the fine-load simulations has published $\beta > 1$. The $\beta = 1$ line is an interpreted topographic threshold, not a modeled avulsion criterion.

Panel 3B compares α_n for $C = 0.0003$, 0.003, and 0.01 at $A = 0.018$ m per flood. The paper reports stronger temporal variability and higher peaks as sediment load increases. The published axis spans approximately 0.1–10 m; exact event timing is not identifiable from the written specification. Panel 3C compares α_{95} across all runs. The main reported pattern is large values for high suspended load and low aggradation, when conveyance occurs through localized breaches. Numerical comparison should report the computed ranges, trend signs, late-time variability, and any failure to preserve this ordering. A high α by itself does not demonstrate avulsion.

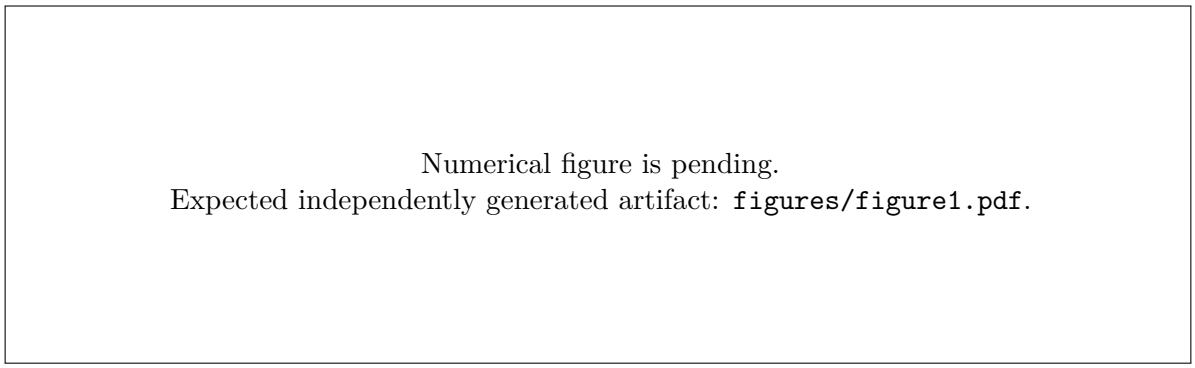


Figure 1: Independent model output for manuscript panels 1A–F: deposit thickness during each listed flood, in meters, on the 48 km by 72 km domain. A–D use $A = 0$, $C = 0.003$, $E = 1$, and the default grain mixture, at floods 61, 251, 501, and 562. E uses $A = 0.036$ and $C = 0.003$ at flood 562; F uses $A = 0$ and $C = 0.0003$ at flood 562. Flow is toward increasing downstream distance. Black marks active channel cells and green marks deposits below 10^{-6} m. The color scale extends to 0.5 m; larger values saturate visually but remain in the saved data and deposit-volume calculation. The solid magenta line in E follows the full active channel–floodplain interface used for the connectivity calculation, not its inundated subset. Snapshots precede each flood's migration and aggradation. These panels test whether levee breaches and abandoned channels focus deposition as the ridge grows. The panel ledger gives the status and commands for each requested result.

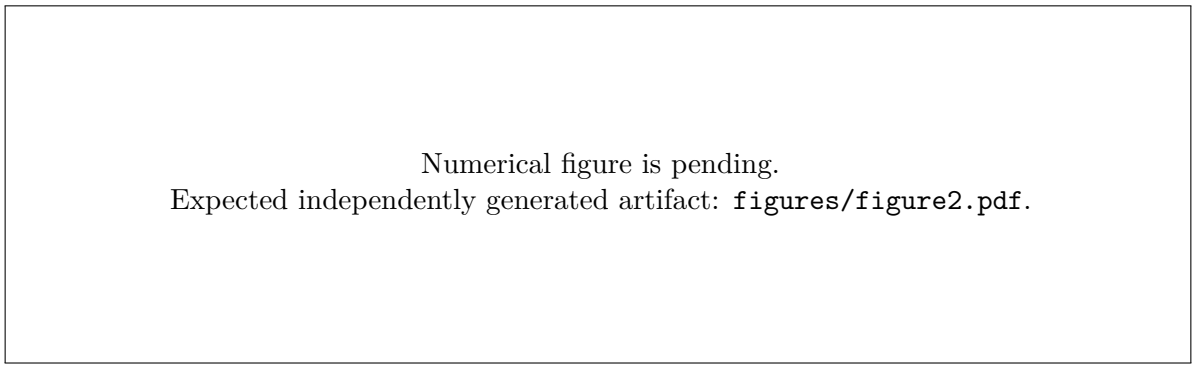


Figure 2: Independent model output for manuscript panels 2A–D: floodplain elevation versus transverse distance at the central grid column, 36.125 km downstream, at the end of floods 50, 225, 400, and 580, before migration and aggradation. Black solid, gray dotted, orange dashed, and red solid lines identify those floods, respectively. Cyan marks saved active channel-bed cells at flood 580. All cases use $E = 1$ and the default grain mixture. A: $A = 0.036$, $C = 0.01$; B: $A = 0.036$, $C = 0.0003$; C: $A = 0$, $C = 0.0003$; D: $A = 0.018$, $C = 0.003$. Aggradation A is in m per flood. Distance and elevation are in km and m, respectively. The initial section is at 3.5875 m; no datum shift is applied. The comparison tests ridge relief growth and channel deepening under different balances of prescribed bed aggradation and computed overbank deposition.

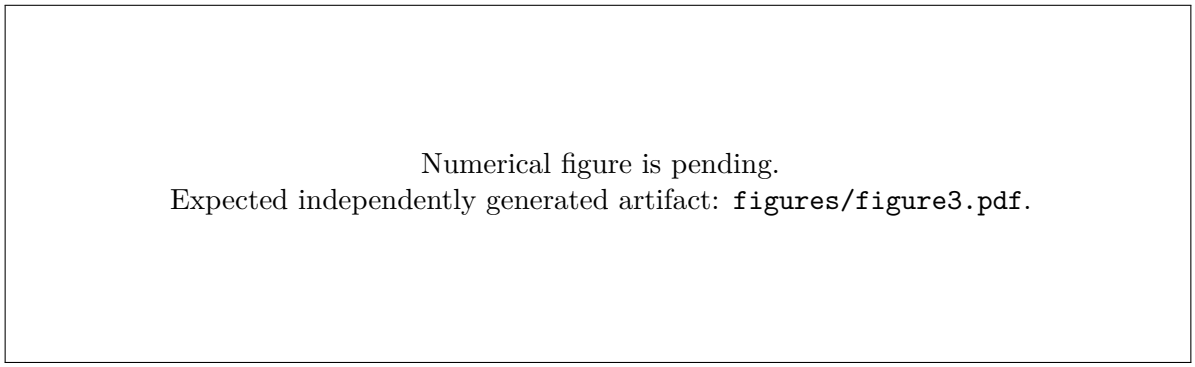


Figure 3: Independent model output for manuscript panels 3A–C. A: arithmetic mean of the per-flood super-elevation ratios β_n over floods 561–580, against imposed bed aggradation A, for all 31 configurations. B: connectivity α at day 12 of each flood for $A = 0.018$ m per flood and $C = 0.0003$, 0.003, and 0.01, with default load and $E = 1$. C: linearly interpolated 95th percentile of α over all 200 floods 381–580 against A. β is dimensionless; α and α_{95} are in meters. Purple, blue, orange, and black identify $C = 0.00003$, 0.0003, 0.003, and 0.01, respectively. Squares denote default load with $E = 1$, diamonds denote fine load, and plus and cross symbols denote $E = 0.5$ and 1.5. Red rings mark altered or unverified settings or incomplete histories. The dashed line at $\beta = 1$ is a published interpretive reference. Missing values create gaps; an incomplete or nonfinite statistical window yields no point. Alpha axes are logarithmic when all available values are positive. The panel ledger records current coverage and provenance.

4.4 Quantitative comparison status

0 of 29 specified numerical comparisons have complete, documented input records. Quantitative manuscript comparison remains pending. No agreement is inferred from missing windows, altered benchmark configurations, or the published axis limits.

The remaining 29 comparisons are pending. Exact required runs and reasons are recorded in `comparison.json`; no pending calculation is assigned a numerical result.

Plot provenance needs regeneration: plot_manifest.json missing; run scripts/plot.py first; metrics.csv missing; deposit_metrics.csv missing.

Published result columns were not read by this comparison. Relative erodibility changes use equally weighted paired cases; alpha variability uses the population coefficient of variation over floods 381–580. Datum comparisons subtract the initial plane and retain the uncertainty of coarse visual estimates.

4.5 Spatial measurements from saved snapshots

The following measurements describe independent output; they do not assign scientific agreement. A dash denotes a missing, noncanonical, inconsistent or unverified requested sample. Diagnostic provenance and all omitted records are retained in the accompanying JSON files.

figures/inundation_metrics.json: diagnostic has not been generated.

figures/planform_metrics.json: diagnostic has not been generated.

Water is sampled at day 12; deposition is sampled at flood end, before migration. Wet area excludes channel ghosts and uses $h > 1$ mm. Mean depth includes dry floodplain cells. The wet-bank fraction uses immediate channel–floodplain faces and both depth and stage tests. Wet patches connect faces sharing a raster vertex; their count is sensitive to threshold rounding. Patch length does not measure discharge.

Panel	Wet area (km^2)	Mean h (m)	Wet banks (fraction)	Patches (count)	Largest patch (km)	Deposit (10^6 m^3)
1A	—	—	—	—	—	—
1B	—	—	—	—	—	—
1C	—	—	—	—	—	—
1D	—	—	—	—	—	—
1E	—	—	—	—	—	—
1F	—	—	—	—	—	—

Deposit volumes include pore space. The next table reports fractions of each event's total bulk deposit volume on its day-12 wet footprint and in previously sampled channel cells that are now floodplain. The latter masks omit unsaved channel positions and do not establish complete abandoned-channel infilling.

Panel	On wet footprint	In earlier channels	Earlier saved floods
1A	—	—	—
1B	—	—	—
1C	—	—	—
1D	—	—	—
1E	—	—	—
1F	—	—	—

Planform sinuosity is centerline arc length divided by the 72 km valley length. Extent and the arc-length-weighted lateral RMS describe the active centerline, not historical channel-belt width. Inverse RMS curvature uses unsmoothed three-point curvatures and is a curvature scale, not a bend wavelength. Zero curvature has no finite inverse scale.

Panel	Sinuosity	Lateral extent (km)	Lateral RMS (km)	Curvature scale (km)
1A	—	—	—	—
1B	—	—	—	—
1C	—	—	—	—
1D	—	—	—	—
1E	—	—	—	—
1F	—	—	—	—

Saved cutoff times and sinuosity series are in `figures/planform_metrics.json`. A count increase between summaries n and $n+1$ dates the remap after flood n. Missing summary intervals remain censored; the last summary cannot date its following remap. The spatial tables supply no unreported numerical target from the publication.

5 Limits of the reconstruction

Assessment of the generated morphology requires the independent maps and profiles alongside their numerical summaries. The automatic comparisons report spatial scales, shape descriptors, trend signs, and variability, but do not establish splay formation, abandoned-channel infilling, ridge concavity, or breach localization. Those claims require direct assessment of the independent maps and profiles and an authored record for each panel. Deposit footprint is not inundation extent. Per-flood alpha components and cutoff counts support temporal comparisons. Day-12 spatial evidence is limited to the requested snapshot floods, and the original initial migration perturbations are undocumented. Completion of all numerical windows therefore does not, by itself, establish scientific reproduction.

The largest identifiable ambiguities concern the sediment concentration units, grain diameters and settling closures, flow resistance, the hydrograph shape, and the dimensional migration scale. These choices change sediment delivery, bankfull capacity, and the time needed to widen the channel belt. Their effects can therefore propagate into every panel. The definitions of ridge height and inundated interface add uncertainty to β and α even when the underlying hydraulic solution is unchanged. Assumption sensitivity must be separated from numerical resolution sensitivity.

The written model fixes channel width and bed aggradation. It does not resolve the bedload supply required to sustain the imposed bed elevation changes. Comparisons consequently concern alluvial ridge construction and connectivity within that idealization. Inferences about natural avulsion likelihood remain bounded by those assumptions.

6 Rebuild and deliverables

Not yet available. A verified clean model-to-report rebuild remains pending. To rebuild this report from the current ledger and figures, run `bash report/build.sh` from the repository root. That command does not run missing simulations and does not establish a completed reconstruction.

A Figure-by-figure record

The record is generated from `docs/panels.csv`. Every numerical panel requires both a data command and a plot command. "Not reproduced" includes unattempted or failed computations; "partially reproduced" must name the missing sampling, physics, resolution, or agreement; "independently reproduced" is reserved for a documented panel whose plotted values come from this implementation. A schematic cannot satisfy a numerical panel.

Panel 1A

Quantity: individual-flood deposit thickness

Configuration: A=0 C=0.003 E=1 default

Sampling: flood 61

Status: not reproduced

Data command:

```
python3 scripts/run_suite.py -runs 11 -floods 580 -data-root data
```

Plot command:

```
python3 scripts/plot.py -data-root data -output figures
```

Documentation: Nicholas et al. (2018), manuscript figure caption; DR1; DR2 configuration columns; docs/implementation.md

Assessment: No requested numerical data are verified for this panel.

Data coverage: current plotted-data verification unavailable. No current plotter coverage record is available.

Data provenance: plot input hashes unavailable; no numerical source files recorded. Coverage claims have not been verified against unchanged numerical inputs.

Current assessment fingerprint: unavailable; see report/generated/panel_evidence.json

Assessment provenance: plot input hashes unavailable; no numerical source files recorded; plot processor/configuration hash missing or changed: scripts/plot.py; plot processor/configuration hash missing or changed: config/runs.csv; comparison processor hash missing or changed; comparison input missing or changed: scripts/compare.py; generated figure missing: figures/figure1.pdf

Source ledger status: not reproduced

Source ledger notes: Coupled implementation and plotting script available; required production outputs pending. Commands are from repository root; see docs/rebuild.md for the clean workflow.

Panel 1B

Quantity: individual-flood deposit thickness

Configuration: A=0 C=0.003 E=1 default

Sampling: flood 251

Status: not reproduced

Data command:

```
python3 scripts/run_suite.py -runs 11 -floods 580 -data-root data
```

Plot command:

```
python3 scripts/plot.py -data-root data -output figures
```

Documentation: Nicholas et al. (2018), manuscript figure caption; DR1; DR2 configuration columns; docs/implementation.md

Assessment: No requested numerical data are verified for this panel.

Data coverage: current plotted-data verification unavailable. No current plotter coverage record is available.

Data provenance: plot input hashes unavailable; no numerical source files recorded. Coverage claims have not been verified against unchanged numerical inputs.

Current assessment fingerprint: unavailable; see report/generated/panel_evidence.json

Assessment provenance: plot input hashes unavailable; no numerical source files recorded; plot processor/configuration hash missing or changed: scripts/plot.py; plot processor/configuration hash missing or changed: config/runs.csv; comparison processor hash missing or changed; comparison input missing or changed: scripts/compare.py; generated figure missing: figures/figure1.pdf

Source ledger status: not reproduced

Source ledger notes: Coupled implementation and plotting script available; required production outputs pending. Commands are from repository root; see docs/rebuild.md for the clean workflow.

Panel 1C

Quantity: individual-flood deposit thickness

Configuration: A=0 C=0.003 E=1 default

Sampling: flood 501

Status: not reproduced

Data command:

```
python3 scripts/run_suite.py -runs 11 -floods 580 -data-root data
```

Plot command:

```
python3 scripts/plot.py -data-root data -output figures
```

Documentation: Nicholas et al. (2018), manuscript figure caption; DR1; DR2 configuration columns; docs/implementation.md

Assessment: No requested numerical data are verified for this panel.

Data coverage: current plotted-data verification unavailable. No current plotter coverage record is available.

Data provenance: plot input hashes unavailable; no numerical source files recorded. Coverage claims have not been verified against unchanged numerical inputs.

Current assessment fingerprint: unavailable; see report/generated/panel_evidence.json

Assessment provenance: plot input hashes unavailable; no numerical source files recorded; plot processor/configuration hash missing or changed: scripts/plot.py; plot processor/configuration hash missing or changed: config/runs.csv; comparison processor hash missing or changed; comparison input

missing or changed: scripts/compare.py; generated figure missing: figures/figure1.pdf

Source ledger status: not reproduced

Source ledger notes: Coupled implementation and plotting script available; required production outputs pending. Commands are from repository root; see docs/rebuild.md for the clean workflow.

Panel 1D

Quantity: individual-flood deposit thickness

Configuration: A=0 C=0.003 E=1 default

Sampling: flood 562

Status: not reproduced

Data command:

```
python3 scripts/run_suite.py -runs 11 -floods 580 -data-root data
```

Plot command:

```
python3 scripts/plot.py -data-root data -output figures
```

Documentation: Nicholas et al. (2018), manuscript figure caption; DR1; DR2 configuration columns; docs/implementation.md

Assessment: No requested numerical data are verified for this panel.

Data coverage: current plotted-data verification unavailable. No current plotter coverage record is available.

Data provenance: plot input hashes unavailable; no numerical source files recorded. Coverage claims have not been verified against unchanged numerical inputs.

Current assessment fingerprint: unavailable; see report/generated/panel_evidence.json

Assessment provenance: plot input hashes unavailable; no numerical source files recorded; plot processor/configuration hash missing or changed: scripts/plot.py; plot processor/configuration hash missing or changed: config/runs.csv; comparison processor hash missing or changed; comparison input missing or changed: scripts/compare.py; generated figure missing: figures/figure1.pdf

Source ledger status: not reproduced

Source ledger notes: Coupled implementation and plotting script available; required production outputs pending. Commands are from repository root; see docs/rebuild.md for the clean workflow.

Panel 1E

Quantity: individual-flood deposit thickness and interface

Configuration: A=0.036 C=0.003 E=1 default

Sampling: flood 562

Status: not reproduced

Data command:

```
python3 scripts/run_suite.py -runs 17 -floods 580 -data-root data
```

Plot command:

```
python3 scripts/plot.py -data-root data -output figures
```

Documentation: Nicholas et al. (2018), manuscript figure caption; DR1; DR2 configuration columns; docs/implementation.md

Assessment: No requested numerical data are verified for this panel.

Data coverage: current plotted-data verification unavailable. No current plotter coverage record is available.

Data provenance: plot input hashes unavailable; no numerical source files recorded. Coverage claims have not been verified against unchanged numerical inputs.

Current assessment fingerprint: unavailable; see report/generated/panel_evidence.json

Assessment provenance: plot input hashes unavailable; no numerical source files recorded; plot processor/configuration hash missing or changed: scripts/plot.py; plot processor/configuration hash missing or changed: config/runs.csv; comparison processor hash missing or changed; comparison input missing or changed: scripts/compare.py; generated figure missing: figures/figure1.pdf

Source ledger status: not reproduced

Source ledger notes: Coupled implementation and plotting script available; required production outputs pending. Commands are from repository root; see docs/rebuild.md for the clean workflow.

Panel 1F

Quantity: individual-flood deposit thickness

Configuration: A=0 C=0.0003 E=1 default

Sampling: flood 562

Status: not reproduced

Data command:

```
python3 scripts/run_suite.py -runs 3 -floods 580 -data-root data
```

Plot command:

```
python3 scripts/plot.py -data-root data -output figures
```

Documentation: Nicholas et al. (2018), manuscript figure caption; DR1; DR2 configuration columns; docs/implementation.md

Assessment: No requested numerical data are verified for this panel.

Data coverage: current plotted-data verification unavailable. No current plotter coverage record is available.

Data provenance: plot input hashes unavailable; no numerical source files recorded. Coverage claims have not been verified against unchanged numerical inputs.

Current assessment fingerprint: unavailable; see report/generated/panel_evidence.json

Assessment provenance: plot input hashes unavailable; no numerical source files recorded; plot processor/configuration hash missing or changed: scripts/plot.py; plot processor/configuration hash missing or changed: config/runs.csv; comparison processor hash missing or changed; comparison input missing or changed: scripts/compare.py; generated figure missing: figures/figure1.pdf

Source ledger status: not reproduced

Source ledger notes: Coupled implementation and plotting script available; required production outputs pending. Commands are from repository root; see docs/rebuild.md for the clean workflow.

Panel 2A

Quantity: mid-domain floodplain cross-section elevation

Configuration: A=0.036 C=0.01 E=1 default

Sampling: floods 50 225 400 580

Status: not reproduced

Data command:

```
python3 scripts/run_suite.py -runs 22 -floods 580 -data-root data
```

Plot command:

```
python3 scripts/plot.py -data-root data -output figures
```

Documentation: Nicholas et al. (2018), manuscript figure caption; DR1; DR2 configuration columns; docs/implementation.md

Assessment: No requested numerical data are verified for this panel.

Data coverage: current plotted-data verification unavailable. No current plotter coverage record is available.

Data provenance: plot input hashes unavailable; no numerical source files recorded. Coverage claims have not been verified against unchanged numerical inputs.

Current assessment fingerprint: unavailable; see report/generated/panel_evidence.json

Assessment provenance: plot input hashes unavailable; no numerical source files recorded; plot processor/configuration hash missing or changed: scripts/plot.py; plot processor/configuration hash missing or changed: config/runs.csv; comparison processor hash missing or changed; comparison input missing or changed: scripts/compare.py; generated figure missing: figures/figure2.pdf

Source ledger status: not reproduced

Source ledger notes: Coupled implementation and plotting script available; required production outputs pending. Commands are from repository root; see docs/rebuild.md for the clean workflow.

Panel 2B

Quantity: mid-domain floodplain cross-section elevation

Configuration: A=0.036 C=0.0003 E=1 default

Sampling: floods 50 225 400 580

Status: not reproduced

Data command:

```
python3 scripts/run_suite.py -runs 10 -floods 580 -data-root data
```

Plot command:

```
python3 scripts/plot.py -data-root data -output figures
```

Documentation: Nicholas et al. (2018), manuscript figure caption; DR1; DR2 configuration columns; docs/implementation.md

Assessment: No requested numerical data are verified for this panel.

Data coverage: current plotted-data verification unavailable. No current plotter coverage record is available.

Data provenance: plot input hashes unavailable; no numerical source files recorded. Coverage claims have not been verified against unchanged numerical inputs.

Current assessment fingerprint: unavailable; see report/generated/panel_evidence.json

Assessment provenance: plot input hashes unavailable; no numerical source files recorded; plot processor/configuration hash missing or changed: scripts/plot.py; plot processor/configuration hash missing or changed: config/runs.csv; comparison processor hash missing or changed; comparison input missing or changed: scripts/compare.py; generated figure missing: figures/figure2.pdf

Source ledger status: not reproduced

Source ledger notes: Coupled implementation and plotting script available; required production outputs pending. Commands are from repository root; see docs/rebuild.md for the clean workflow.

Panel 2C

Quantity: mid-domain floodplain cross-section elevation

Configuration: A=0 C=0.0003 E=1 default

Sampling: floods 50 225 400 580

Status: not reproduced

Data command:

```
python3 scripts/run_suite.py -runs 3 -floods 580 -data-root data
```

Plot command:

```
python3 scripts/plot.py -data-root data -output figures
```

Documentation: Nicholas et al. (2018), manuscript figure caption; DR1; DR2 configuration columns; docs/implementation.md

Assessment: No requested numerical data are verified for this panel.

Data coverage: current plotted-data verification unavailable. No current plotter coverage record is available.

Data provenance: plot input hashes unavailable; no numerical source files recorded. Coverage claims have not been verified against unchanged numerical inputs.

Current assessment fingerprint: unavailable; see report/generated/panel_evidence.json

Assessment provenance: plot input hashes unavailable; no numerical source files recorded; plot processor/configuration hash missing or changed: scripts/plot.py; plot processor/configuration hash missing or changed: config/runs.csv; comparison processor hash missing or changed; comparison input missing or changed: scripts/compare.py; generated figure missing: figures/figure2.pdf

Source ledger status: not reproduced

Source ledger notes: Coupled implementation and plotting script available; required production outputs pending. Commands are from repository root; see docs/rebuild.md for the clean workflow.

Panel 2D

Quantity: mid-domain floodplain cross-section elevation

Configuration: A=0.018 C=0.003 E=1 default

Sampling: floods 50 225 400 580

Status: not reproduced

Data command:

```
python3 scripts/run_suite.py -runs 15 -floods 580 -data-root data
```

Plot command:

```
python3 scripts/plot.py -data-root data -output figures
```

Documentation: Nicholas et al. (2018), manuscript figure caption; DR1; DR2 configuration columns; docs/implementation.md

Assessment: No requested numerical data are verified for this panel.

Data coverage: current plotted-data verification unavailable. No current plotter coverage record is available.

Data provenance: plot input hashes unavailable; no numerical source files recorded. Coverage claims have not been verified against unchanged numerical inputs.

Current assessment fingerprint: unavailable; see report/generated/panel_evidence.json

Assessment provenance: plot input hashes unavailable; no numerical source files recorded; plot processor/configuration hash missing or changed: scripts/plot.py; plot processor/configuration hash missing or changed: config/runs.csv; comparison processor hash missing or changed; comparison input missing or changed: scripts/compare.py; generated figure missing: figures/figure2.pdf

Source ledger status: not reproduced

Source ledger notes: Coupled implementation and plotting script available; required production outputs pending. Commands are from repository root; see docs/rebuild.md for the clean workflow.

Panel 3A

Quantity: beta versus aggradation

Configuration: all 31 configurations

Sampling: last 20 floods

Status: not reproduced

Data command:

```
python3 scripts/run_suite.py -runs all -floods 580 -data-root data
```

Plot command:

```
python3 scripts/plot.py -data-root data -output figures
```

Documentation: Nicholas et al. (2018), manuscript figure caption; DR1; DR2 configuration columns; docs/implementation.md

Assessment: No requested numerical data are verified for this panel.

Data coverage: current plotted-data verification unavailable. No current plotter coverage record is available.

Data provenance: plot input hashes unavailable; no numerical source files recorded. Coverage claims have not been verified against unchanged numerical inputs.

Current assessment fingerprint: unavailable; see report/generated/panel_evidence.json

Assessment provenance: plot input hashes unavailable; no numerical source files recorded; plot processor/configuration hash missing or changed: scripts/plot.py; plot processor/configuration hash missing or changed: config/runs.csv; comparison processor hash missing or changed; comparison input missing or changed: scripts/compare.py; generated figure missing: figures/figure3.pdf

Source ledger status: not reproduced

Source ledger notes: Coupled implementation and plotting script available; required production outputs pending. Commands are from repository root; see docs/rebuild.md for the clean workflow.

Panel 3B

Quantity: alpha time series

Configuration: A=0.018 C=0.0003/0.003/0.01 E=1 default

Sampling: 0.6 duration each flood

Status: not reproduced

Data command:

```
python3 scripts/run_suite.py -runs 7,15,20 -floods 580 -data-root data
```

Plot command:

```
python3 scripts/plot.py -data-root data -output figures
```

Documentation: Nicholas et al. (2018), manuscript figure caption; DR1; DR2 configuration columns; docs/implementation.md

Assessment: No requested numerical data are verified for this panel.

Data coverage: current plotted-data verification unavailable. No current plotter coverage record is available.

Data provenance: plot input hashes unavailable; no numerical source files recorded. Coverage claims have not been verified against unchanged numerical inputs.

Current assessment fingerprint: unavailable; see report/generated/panel_evidence.json

Assessment provenance: plot input hashes unavailable; no numerical source files recorded; plot processor/configuration hash missing or changed: scripts/plot.py; plot processor/configuration hash missing or changed: config/runs.csv; comparison processor hash missing or changed; comparison input missing or changed: scripts/compare.py; generated figure missing: figures/figure3.pdf

Source ledger status: not reproduced

Source ledger notes: Coupled implementation and plotting script available; required production

outputs pending. Commands are from repository root; see docs/rebuild.md for the clean workflow.

Panel 3C

Quantity: alpha95 versus aggradation

Configuration: all 31 configurations

Sampling: last 200 floods

Status: not reproduced

Data command:

```
python3 scripts/run_suite.py -runs all -floods 580 -data-root data
```

Plot command:

```
python3 scripts/plot.py -data-root data -output figures
```

Documentation: Nicholas et al. (2018), manuscript figure caption; DR1; DR2 configuration columns; docs/implementation.md

Assessment: No requested numerical data are verified for this panel.

Data coverage: current plotted-data verification unavailable. No current plotter coverage record is available.

Data provenance: plot input hashes unavailable; no numerical source files recorded. Coverage claims have not been verified against unchanged numerical inputs.

Current assessment fingerprint: unavailable; see report/generated/panel_evidence.json

Assessment provenance: plot input hashes unavailable; no numerical source files recorded; plot processor/configuration hash missing or changed: scripts/plot.py; plot processor/configuration hash missing or changed: config/runs.csv; comparison processor hash missing or changed; comparison input missing or changed: scripts/compare.py; generated figure missing: figures/figure3.pdf

Source ledger status: not reproduced

Source ledger notes: Coupled implementation and plotting script available; required production outputs pending. Commands are from repository root; see docs/rebuild.md for the clean workflow.

B Configuration matrix

This table contains input parameters only. It is generated from `config/runs.csv`, transcribed from the configuration columns of DR2 [2]. A is in m per flood. Units for the published C are unspecified. All runs require 580 floods at the documented resolution.

Run	A	C	E	Load
1	0	3e-05	1	fine
2	0	3e-05	1	default
3	0	0.0003	1	default
4	0.004	0.0003	1	default
5	0.008	0.0003	1	default
6	0.012	0.0003	1	default

Run	A	C	E	Load
7	0.018	0.0003	1	default
8	0.026	0.0003	1	default
9	0.031	0.0003	1	default
10	0.036	0.0003	1	default
11	0	0.003	1	default
12	0.004	0.003	1	default
13	0.008	0.003	1	default
14	0.012	0.003	1	default
15	0.018	0.003	1	default
16	0.026	0.003	1	default
17	0.036	0.003	1	default
18	0.004	0.01	1	default
19	0.012	0.01	1	default
20	0.018	0.01	1	default
21	0.026	0.01	1	default
22	0.036	0.01	1	default
23	0	0.003	1	fine
24	0.008	0.003	1	fine
25	0.018	0.003	1	fine
26	0.036	0.003	1	fine
27	0	0.0003	0.5	default
28	0	0.0003	1.5	default
29	0.012	0.0003	0.5	default
30	0.012	0.0003	1.5	default
31	0	0.0003	1	fine

References

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